# DriveShare

Peer-to-Peer Car Rental Platform

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# What is DriveShare?

## Car Owners

- List cars with price, location & availability
- Manage and edit listings
- Mark cars as available / unavailable
- Confirm or decline booking requests
- Message renters directly

## Renters

- Search cars by location, date & max price
- Watch cars for price drop alerts
- Request and pay for bookings
- Message car owners
- Track booking status in real time

# Program Demo

Or otherwise known as the core user flow

## 01

### Register

Register using email and password, and set up security questions

## 02

### List / Search

Owner lists their car
Renters search for cars they want

## 03

### Watching and Booking

Price change alerts and booking cars by dates

## 04

### Confirm

Owner confirms or declines renter

## 05

### Pay

Simulated payment for confirmation of ride

**Also includes:**

- In-app messaging system between car owner and potential renters
- A Notification system
- Double-booking prevention

# Singleton / Observer

## Singleton Session Managing

### SessionManager

Overrides __new__, returns cached _instance on each call

### get_instance()

Called by auth routes and base template

### Flask-Login

current_user proxy for actual session data

## Observer for Watchlist Managing

### CarSubject

Wraps a Cae and holds list of WatcherObserver

### WatcherObserver

Defines update(car, message)

### RenterWatcher

Writes a notification row to DB in update()

# Mediator / Builder

## Mediator for Notifications

### BaseMediator

Abstract notify(colleague, event, **kwargs)

### NotificationMediator

Routes events to private _handle_* methods

### Booking / Payment /Message

Call mediator.notify(); doesn’t talk to each other

## Builder for Car Listing

### AbstractCarListingBuilder

Defines all set_* methods as abstract

### CarListingBuilder

Chains setters, validates in build() returns dict

### build_car_from_form()

Called by add/edit car routes on form POST

# Proxy / Chain of Responsibility

## Proxy for Payment Proxy

### PaymentService

Abstract pay(amount)

### RealPaymentService

Simulates gateway;
returns fake TXN ID

### PaymentProxy

Validates, then marks booking paid

## Chain of Responsibility for Password Recovery

### SecurityQuestionHandler

set_next() + abstract
handle(user, answers)

### Question1/2/3Handler

Each checks one answer;
passes to next if correct

### PasswordRecoveryChain

Builds a chain from Q1 to Q2 to Q3

# Task Summary

### Auth & Security

User registration | Login / logout | Password hashing | Password recovery

### Car Listings

Add / edit / delete listings | Builder pattern for construction | Toggle availability | Image URL support (Kind of)

### Search & Browse

Filter by location, date range, max price | Double-booking prevention | Car detail page

### Booking System

Request booking | Owner confirm / decline | Cancel booking

### Payment

Simulated payment through Proxy pattern | Status tracking

### Messaging

In-app inbox | Conversations |Mediator pattern for message events

### Watchlist & Alerts

Watch a car with a set  price | Observer pattern triggers notifications on price drop

### Design Patterns

Singleton | Observer | Mediator | Builder | Proxy | Chain of Responsibility

### Frontend

Navy/cobalt CSS theme | Jinja2 templates | Responsive and Intuitive